

VSEPR-SIM THEORETICAL DIVISION
TOP SECRET — Internal Development Document

WO-CHART-FIELD-B

Unified Scientific Data Manifold

*Chart & XChart artifacts / Dynamic event streams /
Flux fields / Radiation transport / Excitation dynamics /
Effective gluon proxy / Qt desktop stack /
Validation mathematics / Universal implementation contract*

VSEPR-SIM Development Work Order Series

Contents

1	Core Thesis and Motivation	3
1.1	The plotting-scale artifact problem	3
1.2	The unified abstraction	3
2	Universal State Vector and Sample Record	4
2.1	Generalised domain	4
2.2	Universal state vector	4
2.3	Primitive sampled record	4
2.4	General table form	4
3	Universal Projection Operator	4
3.1	2D chart	5
3.2	3D surface	5
3.3	Vector-field plot	5
4	Steam Table / Thermodynamic Manifold	5
4.1	State vector	5
4.2	Governing property functions	5
4.3	Thermodynamic identities	5
4.4	Chart projections	6
5	Molecular Dynamics Manifold	6
5.1	Particle state	6
5.2	Force decomposition	6
5.3	Time integration — Velocity Verlet	6
5.4	Chart projections	6
6	Flux Field Manifold	7
6.1	Field types	7
6.2	Conservation form	7
6.3	Discrete flux divergence	7
6.4	Primitive flux record	7
6.5	Chart projections	8
7	Radiation Transport Manifold	8
7.1	Radiation state	8
7.2	Attenuation law	8
7.3	Flux definitions	8
7.4	Dose proxy	8
7.5	Chart projections	8
8	Excitation Dynamics Manifold	9
8.1	Excitation state	9
8.2	Population dynamics	9
8.3	Excitation event struct	9
8.4	Chart projections	9

9 Effective Gluon / Color-Field Proxy Manifold	9
9.1 Effective color-state vector	10
9.2 Color-neutrality residual	10
9.3 Confinement proxy energy	10
9.4 Gluon event struct	10
9.5 Chart projections	10
10 Dynamic Event Stream — .dynx	11
10.1 General event object	11
10.2 Bond-force vector event	11
10.3 .dynx data contract	11
11 File Format Specifications	11
11.1 .chart — lightweight plot recipe	11
11.2 .xchart — rich scientific visualization object	12
11.3 .flux — scalar/vector/tensor field state	13
11.4 .x — universal session bundle	13
12 Qt Desktop Stack	14
12.1 View modes	14
12.2 Desktop opening behaviour	14
12.3 Visualisation families	14
13 Validation Mathematics	15
13.1 Unit consistency	15
13.2 Finite-value gate	15
13.3 Non-negativity gate	15
13.4 Conservation residual	15
13.5 Smoothness gate	15
13.6 Segment discontinuity gate	15
13.7 Axis-offset rejection gate	16
14 Universal Pipeline and Implementation Contract	16
14.1 Universal pipeline	16
14.2 Variable dictionary	16
14.3 Implementation order	17
14.4 Data-in / data-out matrix	18
14.5 Architecture rule	18

§1 Core Thesis and Motivation

1.1 The plotting-scale artifact problem

The previous Python plotting layer produced outputs such as a heat capacity curve for the hydrogen atom with a y-axis offset of the form:

$$\underbrace{1 \times 10^{-10}}_{\text{numerical noise}} + 2.078603 \times 10^1$$

The axis is communicating that $C_p \approx 20.786 \text{ J mol}^{-1} \text{ K}^{-1}$ (the correct monatomic ideal-gas value $\frac{5}{2}R$), but the floating-point residuals in the time-integrated trajectory are being treated as physically meaningful variation and plotted on a collapsed scale. The physics is correct. The representation is not.

This is not a cosmetic problem. A plot that suppresses its own axis scale or offsets it to highlight noise carries no reproducible scientific information. It is missing metadata about:

- what equation and coefficient set produced the curve,
- what phase assumption and species model were used,
- what unit convention the y-axis values are in,
- what validity range the model applies to,
- whether the axis offset was intentional or a numerical artifact.

A PNG image cannot answer any of these questions. The solution is to make the chart itself a structured scientific object that carries its own provenance, validation state, and projection definition alongside the data.

1.2 The unified abstraction

Every output object in VSEPR-SIM is a sampled field over a generalised domain. Steam tables, radiation transport fields, MD trajectories, flux surfaces, excitation timelines, and effective gluon event streams are all projections of the same universal state vector **Q**. One projection operator, all output types. The same validation contract applies to all of them.

File type stack affected by this work order:

Type	Role
<code>.chart</code>	Lightweight plot recipe
<code>.xchart</code>	Rich chart + data + provenance visualization object
<code>.dynx</code>	Dynamic force, bond, vector, excitation, event stream
<code>.flux</code>	Scalar/vector/tensor field state
<code>.x</code>	Universal session / artifact bundle

Work order parts:

Part A Add `.chart` and `.xchart` as first-class artifact formats.

Part B Add Qt desktop support: 2D charts, 3D surfaces, vector fields, scalar field slices, radiation maps, excitation timelines, event trees, batch comparison dashboards.

Part C Connect chart objects to: steam table presolved MD outputs, `.dynx` dynamic event streams, `.flux` field samples, `.x` universal session bundles.

Part D Add validation rules preventing physically meaningless charts from being treated as verified outputs.

§2 Universal State Vector and Sample Record

2.1 Generalised domain

Every sampled object lives on a product domain:

$$\mathcal{D} = \Omega_x \times \Omega_t \times \Omega_\theta \times \Omega_E \times \Omega_s \quad (1)$$

where

$$\Omega_x \subseteq \mathbb{R}^3, \quad \Omega_t \subseteq \mathbb{R}_{\geq 0}, \quad \Omega_\theta \subseteq \mathbb{R}^{n_\theta}, \quad \Omega_E \subseteq \mathbb{R}, \quad \Omega_s \subseteq \mathbb{R}^{n_s}.$$

2.2 Universal state vector

The universal sampled state is:

$$\mathbf{Q} = [\mathbf{x}, t, \boldsymbol{\theta}, E, \mathbf{s}, \mathbf{v}, \mathbf{F}, \boldsymbol{\Phi}, \mathbf{M}, \mathbf{H}] \quad (2)$$

with components:

$\mathbf{x} = (x, y, z)$	spatial position
t	time coordinate
$\boldsymbol{\theta} = (T, P, \rho, \dots)$	thermodynamic / control variables
E	energy coordinate
$\mathbf{s}$	internal state vector
$\mathbf{v} = (v_x, v_y, v_z)$	velocity
$\mathbf{F} = (F_x, F_y, F_z)$	force
$\boldsymbol{\Phi} = (\Phi_0, \Phi_x, \Phi_y, \Phi_z, \Phi_{xx}, \dots)$	field values
$\mathbf{M}$	model metadata
$\mathbf{H}$	provenance hash

2.3 Primitive sampled record

The primitive record for sample i is:

$$\mathcal{R}_i = (\text{id}_i, \mathbf{x}_i, t_i, \boldsymbol{\theta}_i, E_i, \mathbf{s}_i, \mathbf{v}_i, \mathbf{F}_i, \boldsymbol{\Phi}_i, w_i, \mu_i, H_i) \quad (3)$$

where w_i is the statistical weight and μ_i is the uncertainty or variance metric.

2.4 General table form

The full dataset is the table of N records:

$$\mathcal{T} = \{\mathcal{R}_i\}_{i=1}^N \quad (4)$$

In column form, $\mathcal{T}$ is the matrix:

$$\mathcal{T} = [\text{id} \mid x \ y \ z \mid t \mid \theta_1 \cdots \theta_{n_\theta} \mid E \mid s_1 \cdots s_{n_s} \mid v_x \ v_y \ v_z \mid F_x \ F_y \ F_z \mid \Phi_0 \cdots \mid w \mid \mu \mid H] \quad (5)$$

This is the mathematical basis of `.xchart`.

§3 Universal Projection Operator

A chart is a projection of the table $\mathcal{T}$ into a renderable space:

$$\Pi : \mathcal{T} \rightarrow \mathbb{R}^k \quad (6)$$

3.1 2D chart

$$C_{2D} = \{(a_i, b_i)\}_{i=1}^N, \quad a_i = f_a(\mathcal{R}_i), \quad b_i = f_b(\mathcal{R}_i) \quad (7)$$

3.2 3D surface

$$C_{3D} = \{(a_i, b_i, c_i)\}_{i=1}^N, \quad a_i = f_a(\mathcal{R}_i), \quad b_i = f_b(\mathcal{R}_i), \quad c_i = f_c(\mathcal{R}_i) \quad (8)$$

3.3 Vector-field plot

$$C_V = \{(\mathbf{x}_i, \mathbf{u}_i)\}_{i=1}^N, \quad \mathbf{u}_i = f_u(\mathcal{R}_i) \quad (9)$$

Typical choices: $\mathbf{u}_i = \mathbf{v}_i$ (velocity field), $\mathbf{u}_i = \mathbf{F}_i$ (force field), $\mathbf{u}_i = \Phi_{\text{vec},i}$ (flux vector field).

The reason not to build separate plotting systems for steam tables, radiation fields, and gluon events is that the *plot* is just a projection. The data changes. The math does not.

§4 Steam Table / Thermodynamic Manifold

4.1 State vector

$$\mathbf{S}_{\text{steam}} = [T, P, \rho, u, h, s, C_p, C_v, \mu_{\text{visc}}, k, x_q, \phi] \quad (10)$$

Symbol	Quantity	Unit
T	Temperature	K
P	Pressure	Pa
ρ	Density	kg/m ³
u	Specific internal energy	J/kg
h	Specific enthalpy	J/kg
s	Specific entropy	J/(kg K)
C_p	Isobaric heat capacity	J/(mol K)
C_v	Isochoric heat capacity	J/(mol K)
μ_{visc}	Dynamic viscosity	Pa s
k	Thermal conductivity	W/(m K)
x_q	Vapour quality	—
ϕ	Phase code	integer

4.2 Governing property functions

$$\begin{aligned} \rho &= \rho(T, P) & h &= h(T, P) \\ u &= u(T, P) & s &= s(T, P) \\ C_p &= C_p(T, P) & C_v &= C_v(T, P) \end{aligned} \quad (11)$$

4.3 Thermodynamic identities

Enthalpy–internal energy relation:

$$h = u + \frac{P}{\rho} \quad (12)$$

Heat capacities via central finite differences over the MD-derived table:

$$C_p(T_i, P_j) \approx \frac{h(T_{i+1}, P_j) - h(T_{i-1}, P_j)}{T_{i+1} - T_{i-1}} \quad (13)$$

$$C_v(T_i, \rho_j) \approx \frac{u(T_{i+1}, \rho_j) - u(T_{i-1}, \rho_j)}{T_{i+1} - T_{i-1}} \quad (14)$$

Axis-offset artefact note. For a monatomic ideal gas, $C_p = \frac{5}{2}R \approx 20.786 \text{ J mol}^{-1} \text{ K}^{-1}$ is exactly constant. Any trajectory-integrated C_p that varies from this value by less than $\sim 10^{-8} \text{ J mol}^{-1} \text{ K}^{-1}$ is numerical noise. A chart renderer that re-scales the y-axis around these residuals *must* be rejected by the smoothness gate (§13.5) and the chart redrawn with a fixed axis spanning a physically meaningful range.

4.4 Chart projections

$$C_{\text{sat}} = \{(T_i, P_{\text{sat},i})\} \quad \text{2D saturation curve} \quad (15)$$

$$C_h = \{(T_i, P_j, h_{ij})\} \quad \text{3D enthalpy surface} \quad (16)$$

$$C_\rho = \{(T_i, P_j, \rho_{ij})\} \quad \text{3D density surface} \quad (17)$$

$$\epsilon_h(T, P) = h_{\text{MD}}(T, P) - h_{\text{ref}}(T, P) \quad \text{absolute residual} \quad (18)$$

$$\epsilon_{\text{rel},h}(T, P) = \frac{h_{\text{MD}}(T, P) - h_{\text{ref}}(T, P)}{h_{\text{ref}}(T, P)} \quad \text{relative residual} \quad (19)$$

§5 Molecular Dynamics Manifold

5.1 Particle state

$$B_i = [\text{id}_i, \mathbf{x}_i, \mathbf{v}_i, \mathbf{a}_i, m_i, q_i, r_i, \mathbf{s}_i] \quad (20)$$

where $\mathbf{a}_i = \mathbf{F}_i/m_i$ is the acceleration.

5.2 Force decomposition

$$\mathbf{F}_i = \mathbf{F}_{i,\text{bonded}} + \mathbf{F}_{i,\text{near}} + \mathbf{F}_{i,\text{far}} + \mathbf{F}_{i,\text{flux}} + \mathbf{F}_{i,\text{event}} \quad (21)$$

This decomposition is the bead communication spine: each term can be sourced from a different module without restructuring the integrator.

5.3 Time integration — Velocity Verlet

$$\mathbf{x}_{i,n+1} = \mathbf{x}_{i,n} + \mathbf{v}_{i,n} \Delta t + \frac{1}{2} \mathbf{a}_{i,n} \Delta t^2 \quad (22)$$

$$\mathbf{v}_{i,n+1} = \mathbf{v}_{i,n} + \frac{1}{2} (\mathbf{a}_{i,n} + \mathbf{a}_{i,n+1}) \Delta t \quad (23)$$

5.4 Chart projections

Trajectory:

$$C_{\text{traj}} = \{(\mathbf{x}_i(t), y_i(t), z_i(t))\} \quad (24)$$

Energy conservation residual:

$$\epsilon_E(t) = \frac{E(t) - E(0)}{E(0)} \quad (25)$$

Instantaneous kinetic temperature:

$$T(t) = \frac{2}{3Nk_B} \sum_{i=1}^N \frac{1}{2} m_i \|\mathbf{v}_i\|^2 \quad (26)$$

Radial distribution function:

$$g(r) = \frac{1}{4\pi r^2 \rho N} \left\langle \sum_i \sum_{j \neq i} \delta(r - r_{ij}) \right\rangle \quad (27)$$

§6 Flux Field Manifold

6.1 Field types

Scalar flux:

$$\Phi_s = \Phi_s(\mathbf{x}, t, E) \quad (28)$$

Vector flux:

$$\mathbf{\Phi}_v = \begin{pmatrix} \Phi_x \\ \Phi_y \\ \Phi_z \end{pmatrix} = \mathbf{\Phi}_v(\mathbf{x}, t, E) \quad (29)$$

Tensor flux:

$$\mathbf{\Phi}_T = \begin{pmatrix} \Phi_{xx} & \Phi_{xy} & \Phi_{xz} \\ \Phi_{yx} & \Phi_{yy} & \Phi_{yz} \\ \Phi_{zx} & \Phi_{zy} & \Phi_{zz} \end{pmatrix} \quad (30)$$

6.2 Conservation form

$$\frac{\partial q}{\partial t} + \nabla \cdot \mathbf{\Phi} = S \quad (31)$$

where q is the conserved quantity density, $\mathbf{\Phi}$ is the flux, and S is the source/sink term.

6.3 Discrete flux divergence

$$\nabla \cdot \mathbf{\Phi} \approx \frac{\Phi_x(x + \Delta x) - \Phi_x(x - \Delta x)}{2\Delta x} + \frac{\Phi_y(y + \Delta y) - \Phi_y(y - \Delta y)}{2\Delta y} + \frac{\Phi_z(z + \Delta z) - \Phi_z(z - \Delta z)}{2\Delta z} \quad (32)$$

6.4 Primitive flux record

```

1 struct FluxSample {
2     double   x, y, z;           // position (Ang)
3     double   t;                 // time (fs)
4     double   energy_eV;         // energy coordinate
5     double   scalar;            // scalar field value
6     double   vx, vy, vz;        // vector field components
7     double   weight;            // statistical weight
8     uint32_t field_type;         // flux type enum
9     uint64_t source_event_id;    // parent event (0 = none)
10 };

```

Listing 1: Primitive .flux record.

6.5 Chart projections

2D slice at fixed $z = z_0$:

$$C_{\Phi,2D} = \{(x_i, y_i, \Phi_i) : z_i = z_0\} \quad (33)$$

3D vector field:

$$C_{\Phi,\text{vec}} = \{(\mathbf{x}_i, \Phi_i)\} \quad (34)$$

Divergence surface:

$$C_{\nabla \cdot \Phi} = \{(x_i, y_i, (\nabla \cdot \Phi)_i)\} \quad (35)$$

§7 Radiation Transport Manifold

7.1 Radiation state

$$\mathcal{R}_{\text{rad}} = [\mathbf{x}, t, E, \boldsymbol{\Omega}, \Phi, D, \Sigma, \lambda, w] \quad (36)$$

Symbol	Quantity	Unit
$\boldsymbol{\Omega}$	Direction unit vector	—
Φ	Particle flux	$\text{m}^{-2} \text{s}^{-1}$
D	Dose / deposited energy	J/kg (Gy)
Σ	Macroscopic cross section	m^{-1}
$\lambda = 1/\Sigma$	Attenuation length	m
w	Particle weight	—

7.2 Attenuation law

$$I(x) = I_0 e^{-\mu x}, \quad T(x) = \frac{I(x)}{I_0} = e^{-\mu x} \quad (37)$$

7.3 Flux definitions

Integrated flux:

$$\Phi = \frac{dN}{dA dt} \quad (38)$$

Energy-resolved flux:

$$\Phi(E) = \frac{dN}{dA dt dE} \quad (39)$$

7.4 Dose proxy

$$D = \frac{E_{\text{dep}}}{m}, \quad \dot{D} = \frac{dD}{dt} \quad (40)$$

7.5 Chart projections

Energy spectrum, dose map, flux field, and attenuation curve:

$$C_E = \{(E_i, N_i)\} \quad \text{energy spectrum} \quad (41)$$

$$C_D = \{(x_i, y_i, D_i) : z = z_0\} \quad \text{dose map} \quad (42)$$

$$C_{\Phi} = \{(\mathbf{x}_i, \Phi_i)\} \quad \text{flux field} \quad (43)$$

$$C_T = \{(x_i, e^{-\mu x_i})\} \quad \text{attenuation curve} \quad (44)$$

§8 Excitation Dynamics Manifold

8.1 Excitation state

$$\mathcal{X}_i = [\text{id}_i, t_i, \mathbf{x}_i, E_{0,i}, E_{1,i}, \Delta E_i, \sigma_i, p_i, \tau_i] \quad (45)$$

where $\Delta E_i = E_{1,i} - E_{0,i}$, σ_i is the transition cross-section, p_i is the transition probability, and τ_i is the state lifetime.

8.2 Population dynamics

State population vector with normalisation:

$$\mathbf{p}(t) = [p_1(t), p_2(t), \dots, p_n(t)], \quad \sum_{k=1}^n p_k(t) = 1 \quad (46)$$

Transition rate matrix K and population equation:

$$\frac{d\mathbf{p}}{dt} = K \mathbf{p}, \quad \mathbf{p}_{n+1} = \mathbf{p}_n + \Delta t K \mathbf{p}_n \quad (47)$$

8.3 Excitation event struct

```

1 struct ExcitationEvent {
2     uint64_t event_id;
3     uint64_t particle_id;
4     double t; // time (fs)
5     double x, y, z; // position (Ang)
6     double E_initial_eV;
7     double E_final_eV;
8     double delta_E_eV;
9     int state_initial;
10    int state_final;
11    uint32_t cause; // collision / photon_abs / decay /
        thermal / field
12 };

```

Listing 2: Excitation event in .dynx.

8.4 Chart projections

$$C_{\mathcal{X}} = \{(t_i, \Delta E_i)\} \quad \text{transition timeline} \quad (48)$$

$$C_p = \{(t_i, p_k(t_i))\} \quad \text{state population} \quad (49)$$

$$C_{\text{emit}} = \{(\Delta E_i, N_i)\} \quad \text{emission spectrum} \quad (50)$$

§9 Effective Gluon / Color-Field Proxy Manifold

Scope disclaimer. The model in this section is an **effective proxy**, not a first-principles QCD calculation. It does not solve the QCD path integral, does not implement lattice gauge theory, and makes no claim to reproduce the full non-Abelian SU(3) dynamics. It is a reduced representation suitable for event-level bookkeeping in particle cascade simulations and for visualisation of confinement-like phenomena in educational or exploratory contexts. The field `full_qcd = false` must always be set in any `.xchart` or `.dynx` output from this module. The universe already contains enough fraud without compiling more.

9.1 Effective color-state vector

The six-component color-state vector encodes the three color and three anti-color occupancy proxies:

$$\mathbf{G}_i = [g_r, g_g, g_b, \bar{g}_r, \bar{g}_g, \bar{g}_b] \in [0, 1]^6 \quad (51)$$

or, using the reduced scalar for confinement dynamics, the color-excitation state:

$$\chi_c(t) \in [0, 1], \quad \chi_c(t) = 0 \implies \text{color neutral (confined)} \quad (52)$$

9.2 Color-neutrality residual

The residual from exact color neutrality is:

$$\eta_c = \left\| \sum_{i \in \text{hadron}} \mathbf{G}_i \right\|, \quad \eta_c \leq \epsilon_c \implies \text{color-neutral state} \quad (53)$$

9.3 Confinement proxy energy

A phenomenological confinement energy proxy:

$$E_{\text{conf}}(r) = \kappa r + \frac{\alpha_s}{r}, \quad \kappa \approx 0.18 \text{ GeV}^2, \quad \alpha_s \approx 0.1\text{--}0.3 \quad (54)$$

This is the Cornell potential. It is an effective model. It is not derived from the VSEPR-SIM force field and should not be connected to the MD integrator without explicit demarcation.

9.4 Gluon event struct

```

1 struct GluonExcitationEvent {
2     uint64_t event_id;
3     uint64_t parent_particle_id;
4     double t; // time (s)
5     double x, y, z; // position (m)
6     double energy_GeV; // effective energy
7     double confinement_proxy; // Cornell potential value
8     double color_state[6]; // G_i vector
9     uint32_t interaction_type; // emission / absorption /
        annihilation
10    uint64_t state_hash; // provenance
11 };

```

Listing 3: Effective gluon excitation event.

9.5 Chart projections

$$C_\chi = \{(t_i, \chi_{c,i})\} \quad \text{color-excitation timeline} \quad (55)$$

$$C_{\text{conf}} = \{(r_i, E_{\text{conf},i})\} \quad \text{confinement curve} \quad (56)$$

$$C_G = \{(t_i, \mathbf{G}_i)\} \quad \text{color-state history} \quad (57)$$

§10 Dynamic Event Stream — .dynx

10.1 General event object

$$D_i = [\text{id}_i, \text{type}_i, t_i, \mathbf{x}_i, \mathbf{a}_i, \mathbf{b}_i, \mathbf{q}_i, H_i] \quad (58)$$

where $\mathbf{a}_i$ is the primary vector payload, $\mathbf{b}_i$ is the secondary vector payload, and $\mathbf{q}_i$ is the scalar payload vector.

10.2 Bond-force vector event

$$D_{\text{bond}} = [\text{id}, t, \mathbf{x}_a, \mathbf{x}_b, \mathbf{F}_{ab}, \ell, \ell_0, \epsilon_b] \quad (59)$$

Bond length and strain:

$$\ell = \|\mathbf{x}_b - \mathbf{x}_a\| \quad (60)$$

$$\epsilon_b = \frac{\ell - \ell_0}{\ell_0} \quad (61)$$

Bond force (harmonic):

$$\mathbf{F}_{ab} = -k_b (\ell - \ell_0) \frac{\mathbf{x}_b - \mathbf{x}_a}{\ell} \quad (62)$$

Break criteria:

$$|\epsilon_b| > \epsilon_{\text{crit}} \quad \text{or} \quad \|\mathbf{F}_{ab}\| > F_{\text{crit}} \quad (63)$$

10.3 .dynx data contract

Data-in field	Meaning
event_id, event_type	Event identity
time_s	Event time
particle_a, particle_b	Participant IDs
x, y, z	Event position
vx, vy, vz	Velocity at event
Fx, Fy, Fz	Force at event
energy	Event energy
state_initial, state_final	State transition
hash	Event provenance hash

§11 File Format Specifications

11.1 .chart — lightweight plot recipe

A .chart file defines the tuple:

$$C = (S, \Pi, \mathcal{A}, \mathcal{V}) \quad (64)$$

where S is the source dataset path, Π is the projection (column mapping), $\mathcal{A}$ is the axis metadata set, and $\mathcal{V}$ is the validation rule set.

Axis definition:

$$\mathcal{A}_k = (\text{name}_k, \text{unit}_k, f_k) \quad (65)$$

```

1 [chart]
2 version      = "0.2"
3 kind         = "line"
4 title        = "Cp(T) - Hydrogen Atom"
5
6 [source]
7 path         = "h_atom_steam.tsv"
8 x_column     = "T_K"
9 y_column     = "Cp_J_molK"
10
11 [axes]
12 x_label      = "Temperature"
13 x_unit       = "K"
14 x_min        = 100.0
15 x_max        = 6000.0
16 y_label      = "Cp"
17 y_unit       = "J/(mol K)"
18 y_min        = 20.0          # fixed --- prevents offset artifact
19 y_max        = 22.0
20
21 [validation]
22 y_finite     = true
23 y_nonnegative = true
24 smoothness_tol = 0.05
25 axis_offset_check = true    # rejects plots where offset > 1e-6 *
                               range

```

Listing 4: .chart format example.

11.2 .xchart — rich scientific visualization object

An .xchart file defines:

$$\mathcal{X} = (\mathcal{T}, \mathcal{P}, \mathcal{M}, \mathcal{V}, H) \quad (66)$$

where $\mathcal{P} = \{\Pi_1, \Pi_2, \dots, \Pi_k\}$ is the set of chart projections, $\mathcal{M}$ is the metadata block, and H is the provenance hash.

```

1 [xchart]
2 version = "0.2"
3 kind     = "radiation_flux_surface"
4 title    = "Neutron Flux Field Around Source"
5
6 [datasets.primary]
7 format    = "inline_table"
8 columns   = ["x_m", "y_m", "z_m", "time_s", "energy_MeV",
9             "flux_m2_s", "dose_Gy_s"]
10
11 [plot.surface]
12 type     = "3d_surface"
13 x        = "radius_m"
14 y        = "energy_MeV"
15 z        = "flux_m2_s"
16
17 [physics]
18 field     = "neutron_flux"
19 transport_model = "effective_monte_carlo"
20 source    = "decay_event_stream"

```

```

21 units_checked    = true
22
23 [validation]
24 nonneg_flux      = true
25 energy_bins_ok   = true
26 source_hash      = "PLACEHOLDER"

```

Listing 5: .xchart — radiation flux surface.

```

1 [xchart]
2 version = "0.2"
3 kind     = "subatomic_excitation_timeline"
4 title    = "Effective Gluon Excitation Events"
5
6 [datasets.events]
7 format    = "inline_table"
8 columns   = ["event_id", "particle_id", "time_s", "energy_GeV",
9             "confinement_proxy", "color_r", "color_g", "color_b",
10            "anticolor_r", "anticolor_g", "anticolor_b"]
11
12 [plot.timeline]
13 type      = "event_timeline"
14 x         = "time_s"
15 y         = "energy_GeV"
16 group     = "particle_id"
17
18 [plot.phase]
19 type      = "phase_map"
20 x         = "time_s"
21 y         = "confinement_proxy"
22 z         = "energy_GeV"
23
24 [physics]
25 model_class = "effective_subatomic_proxy"
26 full_qcd    = false          # mandatory --- this is NOT a QCD
    solver

```

Listing 6: .xchart — effective gluon excitation timeline.

11.3 .flux — scalar/vector/tensor field state

The .flux file is a binary or text stream of `FluxSample` records (see §6). Text form uses tab-separated columns in the order defined by the file header:

```

1 # flux_version=1
2 # field_type=neutron_flux
3 # units=x:m,t:s,energy:eV,phi:m-2.s-1
4 x_m  y_m  z_m  time_s  energy_eV  phi_scalar  phi_x  phi_y  phi_z
    weight  source_id

```

Listing 7: .flux text header.

11.4 .x — universal session bundle

The bundle is:

$$\mathcal{B} = \{S_{\text{sim}}, \mathcal{T}_{\text{state}}, D_{\text{dyn}}, \mathcal{F}_{\text{flux}}, C_{\text{chart}}, R_{\text{report}}, H_{\mathcal{B}}\} \quad (67)$$

Bundle hash:

$$H_{\mathcal{B}} = \text{Hash}(H_1 \parallel H_2 \parallel \cdots \parallel H_n) \quad (68)$$

Symbol	Contents	Extensions
S_{sim}	Simulation scripts	.vsim
$\mathcal{T}_{\text{state}}$	Particle states	.xyz, .xyza, .xyzf, .xyzc
D_{dyn}	Dynamic event streams	.dynx
$\mathcal{F}_{\text{flux}}$	Field data	.flux
C_{chart}	Charts / visualizations	.chart, .xchart
R_{report}	Reports and logs	.json, .tsv, .md

§12 Qt Desktop Stack

12.1 View modes

The Qt desktop layer treats charts as scientific view modes, not plots. Each view kind maps to a specific projection type from §3.

Qt view class	Projection type	Primary input
Chart2DView	C_{2D}	.chart, .xchart
Chart3DSurfaceView	C_{3D}	.xchart
FieldSliceView	$C_{\Phi, 2D}$	.flux, .xchart
FluxVectorView	$C_{\Phi, \text{vec}}$	.flux
RadiationMapView	C_D, C_{Φ}	.xchart, .dynx
ExcitationTimelineView	$C_{\mathcal{X}}$	.dynx
SubatomicEventView	$C_{\mathcal{X}}, C_G$	.dynx
BatchComparisonView	$\{\Pi_j\}_j$	.x bundle

12.2 Desktop opening behaviour

```

1 vsepr open steam_surface.xchart      # -> Chart3DSurfaceView
2 vsepr open radiation_flux.xchart     # -> RadiationMapView
3 vsepr open gluon_excitation.dynx     # -> SubatomicEventView +
   FluxVectorView
4 vsepr open reactor_batch.x          # -> BatchComparisonView
   dashboard

```

Listing 8: Qt desktop launch examples.

12.3 Visualisation families

Thermodynamic charts. For steam tables and MD-derived properties. Views: 2D saturation curve, 3D property surface, phase map, residual error map, lookup interpolation heatmap.

Radiation charts. For isotope, reactor, shielding, and particle transport. Variables: $\Phi(r, t, E) = dN/(dA dt dE)$. Views: flux vs. radius, dose vs. shielding thickness, energy spectrum, attenuation curve, 3D radiation field.

Flux fields. For thermal, mass, charge, neutron, momentum, and reaction flux. Connects directly to .flux. Views: slice, divergence surface, vector arrows.

Excitation dynamics. For atoms, molecules, radiation events, and reduced quantum states. Views: excitation energy vs. time, transition probability map, emission spectrum, state population histogram, decay cascade timeline.

Effective gluon / subatomic events. Marked `full_qcd = false`. Views: color-state transition timeline, effective confinement energy, subatomic excitation graph, collision event tree, particle cascade map, field intensity surface.

§13 Validation Mathematics

13.1 Unit consistency

Every plotted variable must satisfy dimensional agreement with its declared axis:

$$\dim(f_k) = \dim(\mathcal{A}_k) \quad \forall k \quad (69)$$

Reject if $\dim(f_k) \neq \dim(\mathcal{A}_k)$ for any k .

13.2 Finite-value gate

$$\forall i, k : \text{isfinite}(f_k(\mathcal{R}_i)) = \text{true} \quad (70)$$

Reject if any sample maps to $\{\text{NaN}, +\infty, -\infty\}$.

13.3 Non-negativity gate

For density, absolute temperature, flux magnitude, dose, and mass:

$$q_i \geq 0 \quad \forall i \quad (71)$$

13.4 Conservation residual

For conserved density q governed by (31):

$$\mathcal{R}_q = \frac{\partial q}{\partial t} + \nabla \cdot \Phi - S \quad (72)$$

Accept if $|\mathcal{R}_q| < \epsilon_q$.

13.5 Smoothness gate

Second-difference smoothness check:

$$\Delta^2 y_i = y_{i+1} - 2y_i + y_{i-1} \quad (73)$$

Accept if $|\Delta^2 y_i| < \epsilon_{\text{smooth}}$ unless $\text{boundary}_i = \text{true}$ (phase boundary).

Application to the C_p axis artifact. For a monatomic ideal-gas C_p trace, $|\Delta^2 C_p| \ll 10^{-6} \text{ J mol}^{-1} \text{ K}^{-1}$. A chart renderer that produces an axis offset greater than $10^{-6} \times y_{\text{range}}$ on such a trace must fail this gate and re-render with fixed axis bounds.

13.6 Segment discontinuity gate

At a segment boundary x_b between piecewise model regions:

$$\Delta y_b = |y^+(x_b) - y^-(x_b)| \quad (74)$$

Accept if $\Delta y_b < \epsilon_b$ or $\text{phase_boundary}_b = \text{true}$.

This gate would have caught the ethane/propane plot discontinuities at segment boundaries in the previous implementation.

13.7 Axis-offset rejection gate

Let $y_{\text{range}} = y_{\text{max}} - y_{\text{min}}$ over the plotted data. Define the axis offset as the mantissa displacement introduced by the renderer:

$$\delta_{\text{offset}} = |y_{\text{axis,min}} - y_{\text{min}}| \quad (75)$$

Reject (flag as artifact) if:

$$\frac{\delta_{\text{offset}}}{y_{\text{range}}} < \epsilon_{\text{offset}} \quad \text{and} \quad \delta_{\text{offset}} > 0 \quad (76)$$

Default $\epsilon_{\text{offset}} = 10^{-4}$. A non-zero offset on a nearly-flat curve indicates the renderer is auto-scaling around noise.

§14 Universal Pipeline and Implementation Contract

14.1 Universal pipeline

Every new module must implement the four-stage pipeline:

$$\mathcal{I} \xrightarrow{K} \mathcal{T} \xrightarrow{\Pi_j} C_j \xrightarrow{\mathcal{V}_j} \{\text{pass, fail, } \epsilon\} \xrightarrow{\mathcal{E}} \{.chart, .xchart, \dots\} \quad (77)$$

Input $\mathcal{I}$ Raw simulation or measurement records $\{\mathcal{R}_i^{\text{raw}}\}$.

Transform K Normalise to the universal record schema (3): unit conversion, column mapping, hash computation.

Project Π_j Apply one or more projection operators to produce chart tables C_j .

Validate $\mathcal{V}_j$ Apply all applicable gates from §13. A failing gate produces a diagnostic, not a silent bad chart.

Export $\mathcal{E}$ Write `.chart`, `.xchart`, PNG, SVG, TSV, or bundle into `.x`.

In plain structure:

```

1 raw_data
2   -> normalize_to_universal_table(K)
3   -> for each projection Proj_j:
4       C_j = project(table, Proj_j)
5       result = validate(C_j, rules_j)
6       if result.pass:
7           export(C_j, fmt)
8       else:
9           emit_diagnostic(result)

```

Listing 9: Universal pipeline pseudocode.

14.2 Variable dictionary

The following variable names are canonical across all modules and file formats:

Name	Meaning
id	Persistent sample or particle identifier
x, y, z	Position (Å for MD; m for radiation)
t	Time
T	Temperature (K)
P	Pressure (Pa)
rho	Density (kg/m ³)
u	Specific internal energy (J/kg)
h	Specific enthalpy (J/kg)
s	Specific entropy (J/(kg K))
Cp, Cv	Heat capacities (J/(mol K))
mu	Dynamic viscosity (Pa s)
k	Thermal conductivity (W/(m K))
E	Energy (eV for MD; MeV/GeV for particle)
vx, vy, vz	Velocity (Å/fs)
Fx, Fy, Fz	Force (eV/Å)
Phi	Scalar flux
Phi_x, Phi_y, Phi_z	Vector flux components
D	Dose (Gy)
Omega	Radiation direction unit vector
Sigma	Macroscopic cross section (m ⁻¹)
state_i, state_j	Initial and final excitation state indices
delta_E	Transition energy
c	Color-state proxy vector $\mathbf{G}_i$
eta_c	Color-neutrality residual
E_conf	Confinement proxy energy
w	Sample weight
epsilon	Validation residual
H	Provenance hash

14.3 Implementation order

1. Implement `.chart` format parser and writer.
2. Implement `.xchart` format with embedded table support.
3. Implement `.flux` binary and text record I/O.
4. Connect `.xchart` to `.flux` (field visualization).
5. Connect `.xchart` to `.dynx` (event timeline visualization).
6. Add Qt `Chart2DView` and `Chart3DSurfaceView`.
7. Add Qt `FluxVectorView` and `FieldSliceView`.
8. Add Qt `ExcitationTimelineView`.
9. Add Qt `RadiationMapView` and `SubatomicEventView`.
10. Add Qt `BatchComparisonView`.
11. Add all validation gates (§13).
12. Bundle all outputs through `.x`.

14.4 Data-in / data-out matrix

Module	Data-in	Data-out
Steam MD table	$T, P, \rho, u, h, s, C_p, C_v, \mu, k$	<code>.xchart</code> , lookup cache, phase maps
Molecular dynamics	$\mathbf{x}, \mathbf{v}, m, q, \mathbf{F}$, forcefield	<code>.xyzf</code> , <code>.dynx</code> , <code>.xchart</code>
Flux field	$\mathbf{x}, t, E, \Phi, \mathbf{\Phi}$	<code>.flux</code> , vector/slice charts
Radiation transport	particle type, $E, \mathbf{\Omega}, \Sigma, \Phi, D$	<code>.dynx</code> , <code>.flux</code> , dose maps
Excitation dynamics	$E_i, E_j, \Delta E, p, \tau$	timelines, emission spectra
Effective gluon proxy	$\mathbf{G}, E_{\text{conf}}, \chi_c$	color-state maps, event trees
Chart engine	source columns, axes, units, rules	PNG/SVG/Qt view
XChart engine	embedded datasets, projections, hashes	multi-view scientific artifact
Bundle <code>.x</code>	all artifact types	replayable session archive

14.5 Architecture rule

Do not make gluons, radiation, flux, and steam tables separate isolated plotting systems. They are all projections of the universal state vector $\mathbf{Q}$ (2) onto appropriate subspaces:

$$\mathcal{D} = \{X, Y, Z, T, E, S, V, M, H\} \quad (78)$$

where X, Y, Z are position; T is time or temperature; E is energy; S are state variables; V are vector field components; M is model metadata; and H is the provenance hash. Then `.xchart` maps those fields into views. One spine. All output types fall from it.

WO-CHART-FIELD-B — VSEPR-SIM Theoretical Division. One abstraction. All the physics. Someone alert the academy before they ruin it with a subcommittee.